Chapter 25: Measuring the Cost of Living

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Learning Objectives

25.1 Calculate the consumer price index (CPI) and relate CPI to inflation.

25.2 Explain why the CPI is an imperfect measure of the cost of living.

25.3 Define the CPI and the GDP Deflator and describe how they measure the overall price level.

25.4 Convert dollar figures from different times using price indexes.

25.5 Describe the relationship between the nominal interest rate, inflation, and the real interest rate.

Key Points

- The consumer price index (CPI) shows the cost of a basket of goods and services relative to the cost of the same basket in the base year. The index is used to measure the overall level of prices in the economy. The percentage change in the CPI measures the inflation rate.
- The CPI is an imperfect measure of the cost of living for three reasons. First, it does not take into account consumers' ability to substitute toward goods that become relatively cheaper over time. Second, it does not allow for increases in the purchasing power of the dollar due to the introduction of new goods. Third, it is distorted by unmeasured changes in the quality of goods and services. Because of these measurement problems, the CPI overstates true inflation.
- Like the CPI, the GDP deflator measures the overall level of prices in the economy. The two price indexes usually move together, but there are important differences. Unlike the CPI, the GDP deflator reflects the prices of goods and services produced domestically rather than of those bought by consumers. As a result, imported goods affect the CPI but not the GDP deflator. In addition, while the CPI uses a fixed basket of goods, the GDP deflator automatically changes the group of goods and services over time as the composition of GDP changes.

Key Points

- Dollar figures from different times do not represent a valid comparison of purchasing power. To compare a dollar figure from the past to a dollar figure today, the older figure should be inflated using a price index.
- Various laws and private contracts use price indexes to correct for the effects of inflation. The tax laws, however, are only partially indexed for inflation.
- Correcting for inflation is especially important when looking at interest rates. The nominal interest rate is the interest rate usually reported; it is the rate at which the number of dollars in a savings account increases over time. By contrast, the real interest rate is the rate at which the purchasing power of a savings account increases (or decreases) over time. The real interest rate equals the nominal interest rate minus the rate of inflation.

The Consumer Price Index (CPI)

- Measures the typical consumer’s cost of living
- The basis of cost-of-living adjustments (COLAs) in many contracts and in Social Security
- The CPI measures consumption, so items like stock, bonds and real estate are not included as those are viewed as investment.
- The Bureau of Labor Statistics is the US’ agency that reports the CPI on a monthly basis.

How the CPI Is Calculated

- **Fix the "basket."**
  The Bureau of Labor Statistics (BLS) surveys consumers to determine what's in the typical consumer's "shopping basket."
  - This consists of all goods and services purchased for consumption by the reference population.
  - There are over 200 categories of goods that are aggregated up to 8 major groups: **food and beverages, housing, apparel, transportation, medical care, recreation, education and communication,** and **other goods and services**
  - About 80,000 items are recorded
  - Any changes in the item (quality/quantity) are also collected

How the CPI Is Calculated

- **Find the prices.**
  - The BLS collects data on the prices of all the goods in the basket.
  - To collect the data, BLS data collectors either visit or call retail stores, service establishments, retail units and doctors’ offices all over the US.
  - Each item is denoted with a pricing period (roughly 10 days)
- **Compute the basket’s cost.**
  - Use the prices to compute the total cost of the basket.
- **Taxes**
  - Sales and excise taxes are included in the CPI
  - Income and social security taxes are not

How the CPI Is Calculated

- **Choose a base year and compute the index.**
  The CPI in any year equals

$$100 \times \frac{\text{cost of basket in current year}}{\text{cost of basket in base year}}$$

- **Compute the inflation rate.**
  The percentage change in the CPI from the preceding period.

$$\text{Inflation rate} = \frac{\text{CPI this year} - \text{CPI last year}}{\text{CPI last year}} \times 100\%$$

CPI Calculation Example

year	*price of mask*	*price of toilet paper*	*cost of basket*
2018	\$5	\$2.00	\$5 x 2 + \$2 x 4 = \$18
2019	\$5	\$2.25	\$5 x 2 + \$2.25 x 4 = \$19
2020	\$10	\$3.00	\$10 x 2 + \$3 x 4 = \$32

basket:

- 2 masks
- 4 rolls

Compute CPI in each year using 2018 base year:

2018: 100 x (\$18/\$18) = 100

2019: 100 x (\$19/\$18) = 105.6

2020: 100 x (\$32/\$18) = 177.8

Inflation rate:

$$= \frac{105.6 - 100}{100} \times 100\% = 5.6\%$$

$$= \frac{177.8 - 105.6}{105.6} \times 100\% = 68.4\%$$

Activity: Calculate the CPI

- CPI basket:
  {5 lbs beef, 5 lbs chicken}
- The CPI basket cost $100 in 2018, the base year.

	price of beef	price of chicken
2018	$10	$10
2019	$12	$13
2020	$16	$25

1. Compute the CPI in 2019.
2. What was the CPI inflation rate from 2018–2020?

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Activity: Calculate the CPI

CPI basket:
{5 lbs beef,5 lbs chicken}

The CPI basket cost $100 in 2018, the base year.

	price of beef	price of chicken
2018	$10	$10
2019	$12	$13
2020	$16	$25

1. Compute the CPI in 2019:

Cost of CPI basket in 2019
= ($12 x 5) + ($13 x 5) = $125

CPI in 2019 = 100 x ($125/$100) = 125

Activity: Calculate the CPI

- CPI basket:
  {5 lbs beef,5 lbs chicken}

- The CPI basket cost $100 in 2018, the base year.

	price of beef	price of chicken
2018	$10	$10
2019	$12	$13
2020	$16	$25

1. What was the inflation rate from 2018–2020?

Cost of CPI basket in 2020

= ($16 x 5) + ($25 x 5) = $205

CPI in 2020 = 100 x ($205/$100) = 205

CPI inflation rate = (205 – 100)/100 = 105%

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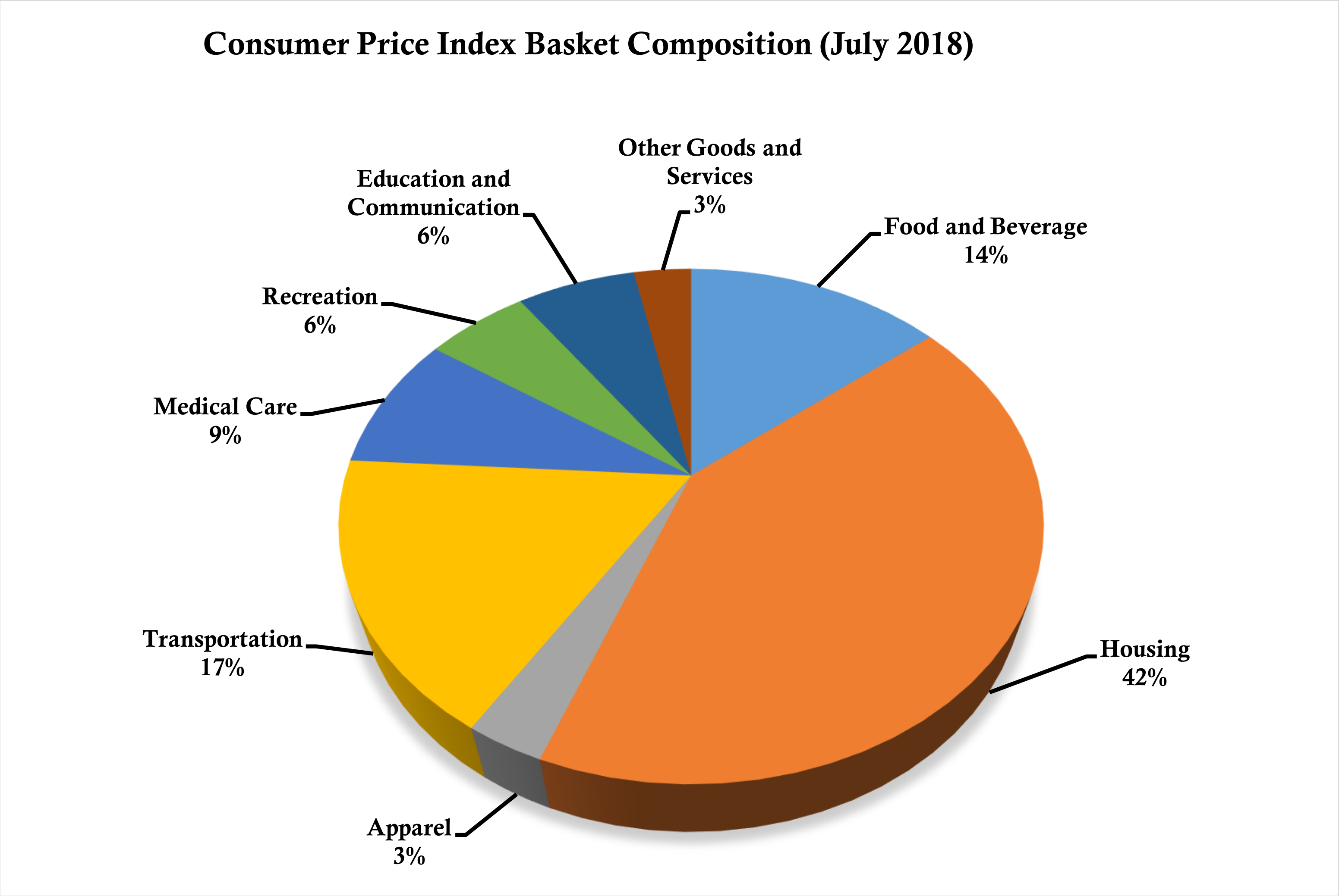
What’s in the CPI’s Basket?



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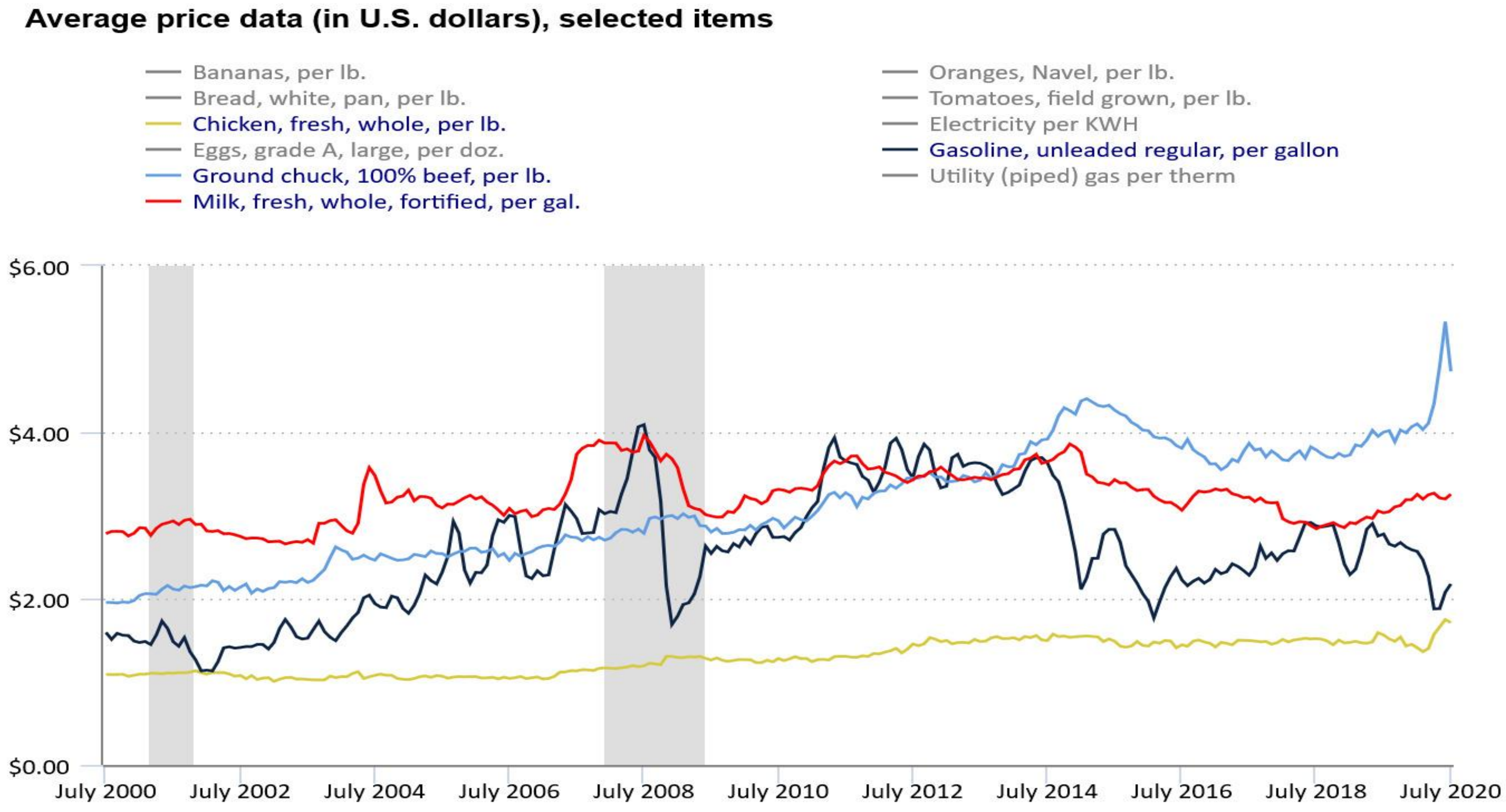
Average Price Data of Select Goods



Hover over chart to view data.
Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.
Source: U.S. Bureau of Labor Statistics.

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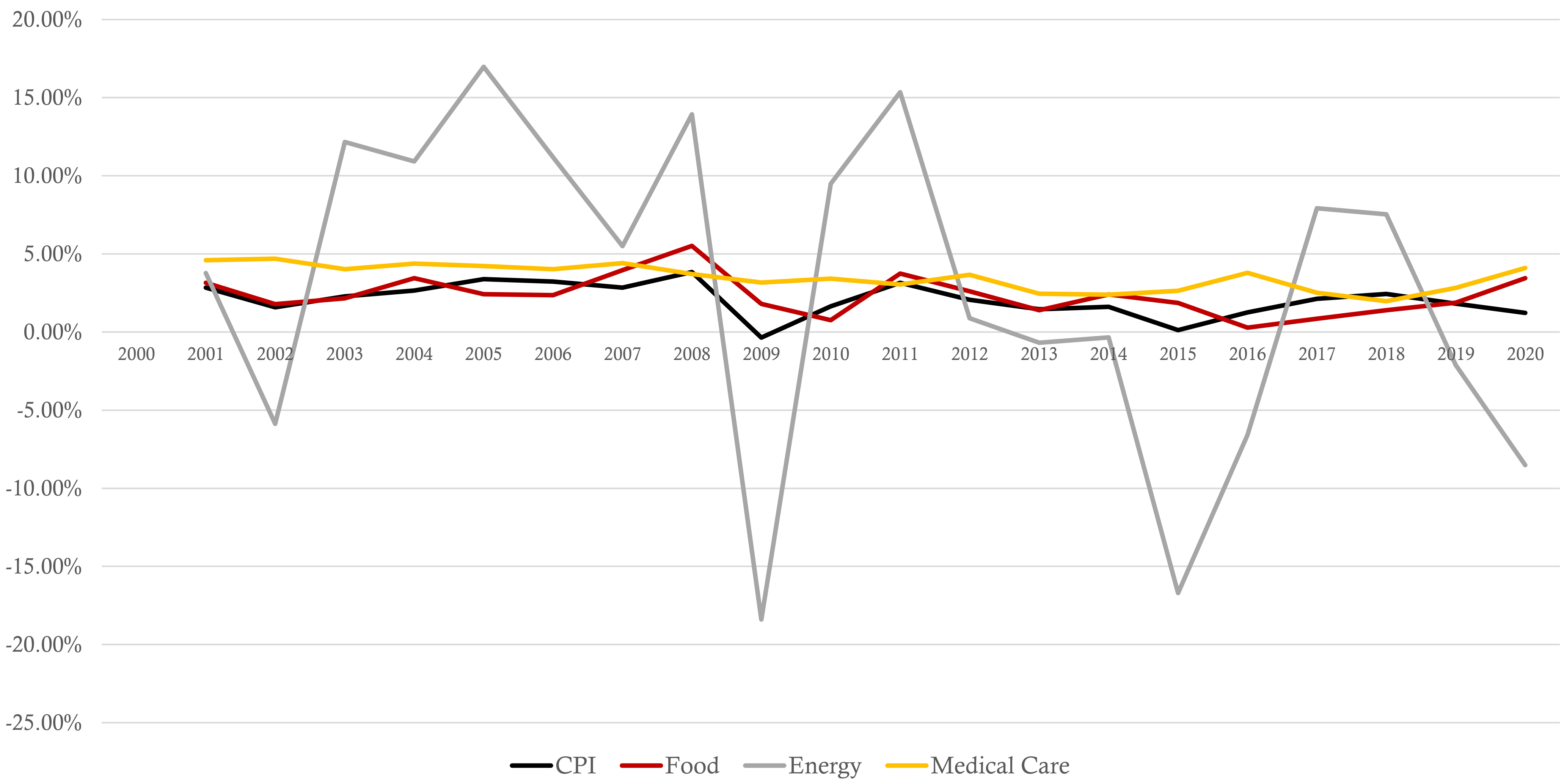
Inflation Rates of Various Categories



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Problems with the CPI: Substitution Bias

- Over time, some prices rise faster than others.
- Consumers substitute toward goods that become relatively cheaper, mitigating the effects of price increases.
- The CPI misses this substitution because it uses a fixed basket of goods.
- Thus, the CPI overstates increases in the cost of living.

Substitution Bias Example

CPI basket:
{5lbs beef, 5lbs chicken}

In 2018 and 2019,
households bought CPI basket.

	beef	chicken	cost of CPI basket
2018	\$10	\$10	\$100
2019	\$12	\$13	\$125
2020	\$16	\$25	\$205

In 2020, households bought {8 lbs beef, 2 lbs chicken}.

1. Compute cost of the 2020 household basket.
2. Compute % increase in cost of household basket over 2019–20, compare to CPI inflation rate.

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Substitution Bias: Answers

CPI basket:
{5lbs beef, 5lbs chicken}

In 2018 and 2019, households bought CPI basket.

	beef	*chicken*	*cost of CPI basket*
2018	$10	$10	$100
2019	$12	$13	$125
2020	$16	$25	$205

In 2020, households bought {8 lbs beef, 2 lbs chicken}.

1. Compute cost of the 2020 household basket.

   ($16 x 8) + ($25 x 2) = $178

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Substitution Bias: Answers

CPI basket:
{5lbs beef, 5lbs chicken}

In 2018 and 2019, households bought CPI basket.

	beef	*chicken*	*cost of CPI basket*
2018	$10	$10	$100
2019	$12	$13	$125
2020	$16	$25	$205

In 2020, households bought {8 lbs beef, 2 lbs chicken}.

1. Compute % increase in cost of household basket over 2019–20, compare to CPI inflation rate.

   Rate of increase: ($178 – $125)/$125 = 42.4%

   CPI inflation rate from previous problem = 105%

Problems with the CPI: Introduction of New Goods

- The introduction of new goods increases variety, allows consumers to find products that more closely meet their needs.
- In effect, dollars become more valuable.
- The CPI misses this effect because it uses a fixed basket of goods.
- Thus, the CPI overstates increases in the cost of living.

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Problems with the CPI: Unmeasured Quality Change

- Improvements in the quality of goods in the basket increase the value of each dollar.
- The BLS tries to account for quality changes but probably misses some, as quality is hard to measure.
- Thus, the CPI overstates increases in the cost of living.

Old Flat Panel Tube TV

OLED TV

Problems with the CPI: Technical Adjustments

- The BLS collects over a million prices a year and does have a sampling error.
- In a given month the standard error a one-month price change is about 0.03%.
- This means that if inflation increases by 0.5% in a month, then, at the 95% confidence level, inflation had a range of 0.44% to 0.56% (two SDs away from the mean).
- Each of these problems causes the CPI to overstate cost of living increases.
- The academic community called for these initial criticisms and a commission was formed to find out the true overstatement. The Boskin report (Final Report of the Advisory Commission to Study the Consumer Price Index) was the result.
- The study found that the CPI overstated inflation by about 1.1-1.3%
- The BLS has made technical adjustments after the report (new upward bias of about 0.3%), but now some think it introduced a downward bias.
- This is important because Social Security payments and many contracts have COLAs tied to the CPI.

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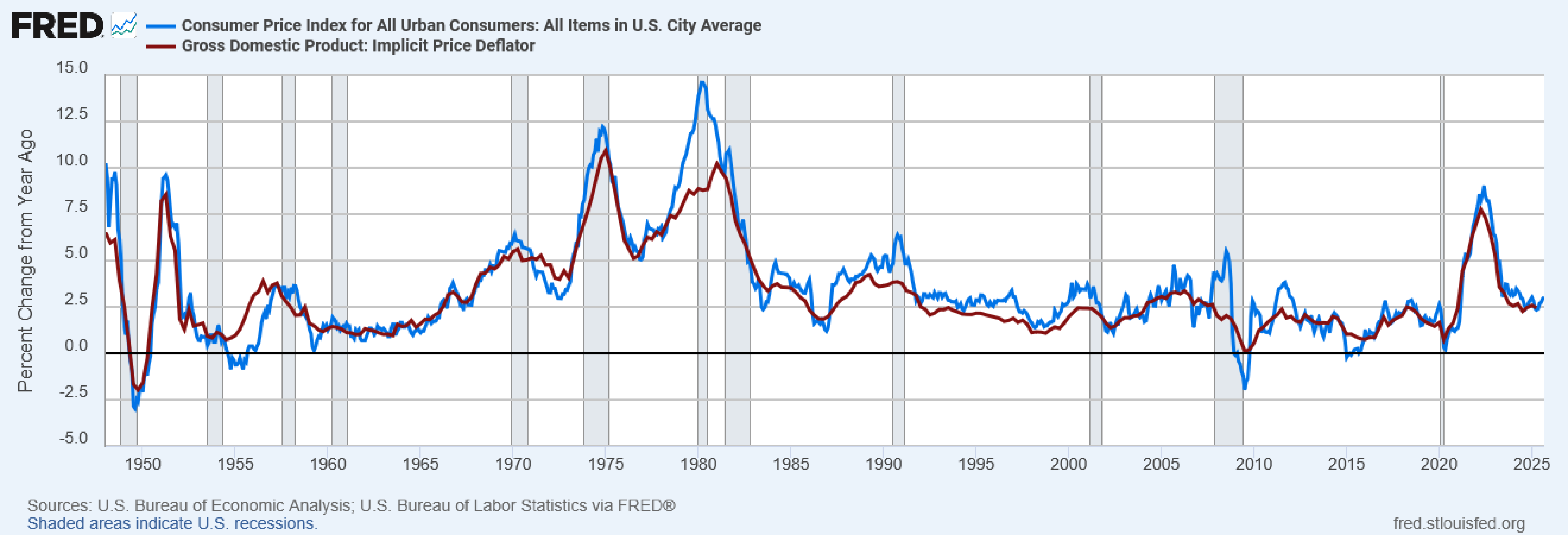
The CPI and the GDP Price Deflator (1948 – 2020)



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Contrasting the CPI and GDP Deflator

Imported consumer goods:

- included in CPI
- excluded from GDP deflator

Capital goods:

- excluded from CPI
- included in GDP deflator (if produced domestically)

The basket:

- CPI uses fixed basket
- GDP deflator uses basket of currently produced goods & services

This matters if different prices are changing by different amounts.

Correcting Variables for Inflation: Comparing Dollar Figures from Different Times

- Inflation makes it harder to compare dollar amounts from different times.

- Example: the minimum wage
  - \$1.15 in Dec 1964
  - \$7.25 in February 2021
- Did min wage have more purchasing power in Dec 1964 or February 2021?
- To compare, use CPI to convert 1964 figure into "2021 dollars"…

Correcting Variables for Inflation: Comparing Dollar Figures from Different Times

$$\text{Amount in today's dollars} = \text{Amount in year } T \text{ dollars} \times \frac{\text{Price level today}}{\text{Price level in year } T}$$

- In our example,
  - "year *T*" is 1964, "today" is 2021
  - Min wage was $1.15 in year *T*
  - CPI = 31.3 in year *T*, CPI = 263.16 today

The minimum wage in 1964 was $9.67 in 2021 dollars.

$$\$9.67 = \$1.15 \times \frac{263.16}{31.3}$$

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Keep It Reel: Movie Prices and Inflation

- How do we adjust movie ticket prices from one year to the next?

Correcting Variables for Inflation: Comparing Dollar Figures from Different Times

$$\text{Amount in year } T \text{ dollars} = \text{Amount in today's dollars} \times \frac{\text{Price level in year } T}{\text{Price level today}}$$

- Example: Gas prices in 1975
  - "year *T*" is 1975, "today" is 2021
  - The price of gas was \$0.54 in year *T*
  - CPI = 53.8 in year *T*, CPI = 263.16 today

Gas Prices in 1975 would be \$0.54 in 2021 dollars.

$$\$0.54 = \$2.65 \times \frac{53.8}{259.1}$$

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Correcting Variables for Inflation

- What is the highest grossing film of all time?

Rank	Title	Year	Adjusted Gross	Unadjusted Gross
1	Gone with the Wind	1939	$1,850,581,586	$200,882,193
2	Star Wars Episode IV: A New Hope	1977	$1,629,496,559	$460,998,007
3	The Sound of Music	1965	$1,303,502,105	$160,888,776
4	E.T.: The Extra-Terrestrial	1982	$1,297,730,421	$439,454,989
5	Titanic	1997	$1,240,054,754	$674,354,882
6	The Ten Commandments	1956	$1,198,431,667	$65,500,000
7	Jaws	1975	$1,172,447,655	$280,083,300
8	Doctor Zhivago	1965	$1,135,632,932	$111,721,910
9	The Exorcist	1973	$1,011,798,348	$233,005,644
10	Snow White and the Seven Dwarfs	1937	$997,168,333	$184,925,486

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Activity: Comparing Tuition Increases

Tuition and Fees at U.S. Colleges and Universities		
	1990	2010
Private non-profit 4-year	$9,340	$27,293
Public 4-year	$1,908	$7,605
Public 2-year	$906	$2,713
CPI	130.7	218.1

Instructions:

1. Express the 1990 tuition figures in 2010 dollars, then compute the percentage increase for all three types of schools.
2. Which type experienced the largest increase in real tuition costs?

Activity: Comparing Tuition Increases

1. Express the 1990 tuition figures in 2010 dollars, then compute the percentage increase for all three types of schools.
   1. Private non-profit 4-year (2010 $): [(218.1/130.7) x $9,340] = $15,586
   2. Public 4-year (2010 $): [(218.1/130.7) x $1,908] = $3,184
   3. Public 2-year (2010 $): [(218.1/130.7) x $906] = $1,512

2. Which type experienced the largest increase in real tuition costs?
   1. (($27,293 - $15,586) / $15,586) x 100 = 75.1%
   2. (($7,605 - $3,184) / $3,184) x 100 = 138.9%
   3. (($2,713 - $1,512) / $1,512) x 100 = 79.4%

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Correcting Variables for Inflation: Indexation

A dollar amount is **indexed** for inflation if it is automatically corrected for inflation by law or in a contract.

For example, the increase in the CPI automatically determines

- the COLA in many multi-year labor contracts
- adjustments in Social Security payments and federal income tax brackets

Correcting Variables for Inflation: Real vs. Nominal Interest Rates

- The nominal interest rate:
  - the interest rate not corrected for inflation
  - the rate of growth in the dollar value of a deposit or debt
- The real interest rate:
  - corrected for inflation
  - the rate of growth in the purchasing power of a deposit or debt
- Real interest rate = (nominal interest rate) – (inflation rate)
    - Or: $r = n - i$

Correcting Variables for Inflation: Real vs. Nominal Interest Rates

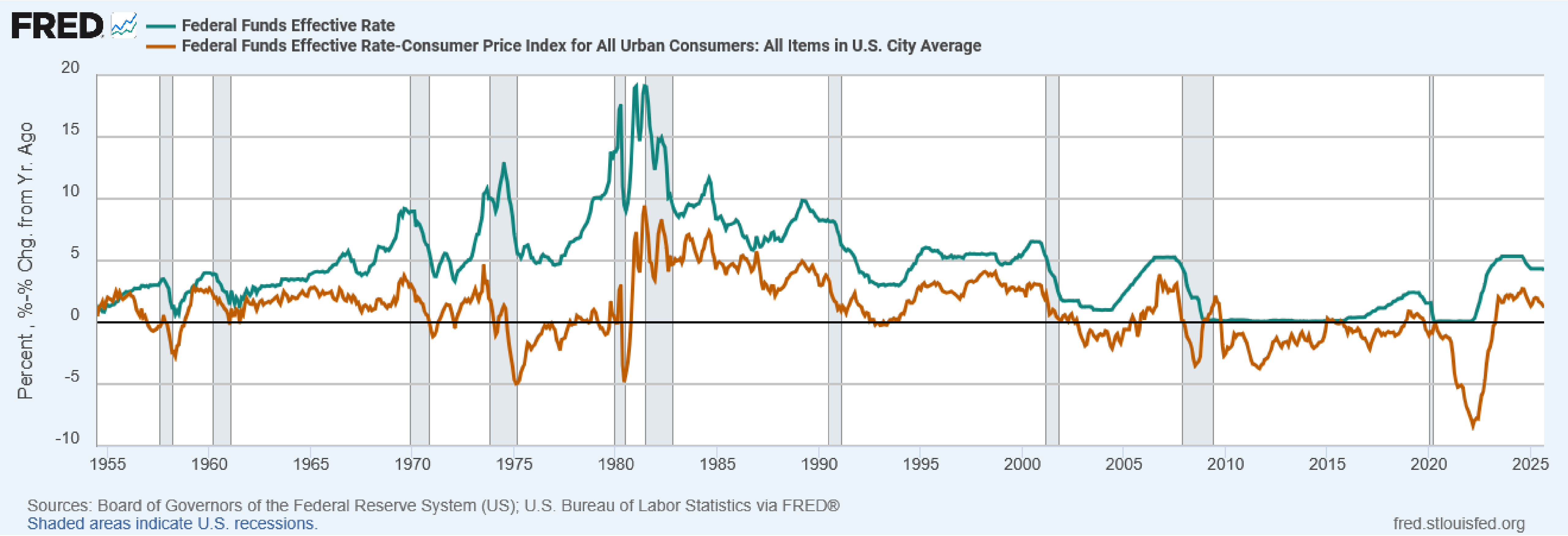
Example:

- Deposit $1,000 for one year.
- Nominal interest rate is 9%.
- During that year, inflation is 3.5%.
- Real interest rate

  = Nominal interest rate – Inflation

  = 9.0% – 3.5% = 5.5%
- The purchasing power of the $1000 deposit has grown 5.5%.

Nominal and Real Interest Rates in the U.S.,1954–2025



Key Takeaways

- **CPI as a Measure:** The Consumer Price Index (CPI) measures the cost of a typical consumer's basket relative to a base year, and is used to gauge inflation.
- **Imperfect Measure:** The CPI overstates inflation because it doesn't fully account for substitution, new goods, and quality improvements.
- **Comparing Price Indexes:** The CPI includes imported goods, while the GDP deflator reflects domestically produced goods and services.
- **Adjusting for Inflation:** Price indexes allow economists to compare dollar values across time, revealing real (inflation-adjusted) purchasing power.
- **Interest Rates:** The real interest rate equals the nominal rate minus inflation, showing how purchasing power changes over time.

Knowledge Check

1. What does the CPI measure?
2. Name two things the CPI tends to overstate in terms of inflation?
3. How does the GDP deflator differ from the CPI?
4. If nominal interest rates are 8% and inflation is 3%, what is the real interest rate?
5. When comparing dollar amounts from different years, what adjustment should be made?

Knowledge Check Answers

1. The CPI measures the cost of a fixed basket of goods and services purchased by consumers.
2. The CPI tends to overstate **inflation** because it does not fully account for **substitution bias** and **quality improvements** (it also misses the effect of new goods).
3. The GDP deflator measures prices of goods and services produced domestically.
4. Real interest rate = 8% – 3% = **5%**.
5. Convert values to a common year's dollars using the CPI for meaningful comparison.